CHAPTER EXERCISE: Answers for this chapter start on page 280.

A calculator should NOT be used on the following questions.

1

Tim's diet plan calls for 60 grams of protein per day. If Tim were to meet this requirement by only eating a certain protein bar that contains 30 grams of protein, how many protein bars would he have to buy to last a week?

2

An electronics company sells computer monitors and releases a new model every year. With each new model, the company increases the screen size by a constant amount. In 2005, the screen size was 15.5 inches. In 2011, the screen size was 18.5 inches. Which of the following best describes how the screen size changed between 2005 and 2011?

A) The company increases the screen size by 0.5 inch every year.

B) The company increases the screen size by 1 inches every year.

C) The company increases the screen size by 2 inches every year.

D) The company increases the screen size by 3 inches every year.

3

As a submarine descends into the deep ocean, the pressure it must withstand increases. At an altitude of -700 meters, the pressure is 50 atm (atmospheres), and at an altitude of -900 meters, the pressure is 70 atm. For every 10 meters the submarine descends, the pressure it faces increases by n, where n is a constant. What is the value of n ?

A) 0.1

B) 1

C) 2

D) 10

4

An empty pool can be filled in 5 hours if water is pumped in at 300 gallons an hour. How many hours would it take to fill the pool if water is pumped in at 500 gallons an hour?

5

If a apples cost d dollars, which of the following expressions gives the cost of 20 apples, in dollars?

A) $\frac{20a}{d}$

B) $\frac{20d}{a}$

C) $\frac{a}{20d}$

D) $\frac{20}{ad}$

6

During a race on a circular race track, a racecar burns fuel at a constant rate. After lap 4, the racecar has 22 gallons left in its tank. After lap 7, the racecar has 18 gallons left in its tank. Assuming the racecar does not refuel, after which lap will the racecar have 6 gallons left in its tank?

A) Lap 13
B) Lap 15
C) Lap 16
D) Lap 19

7

By 1:00 PM, a total of 40 boxes had been unloaded from a delivery truck. By 3:30 PM, a total of 65 boxes had been unloaded from the same truck. If boxes are unloaded from the truck at a constant rate, what is the total number of boxes that will have been unloaded from the truck by 7:00 PM?

A calculator is allowed on the following questions.

8

A rolling ball covers a distance of 2400 feet in 4 minutes. What is the ball's average speed, in inches per second? (12 inches = 1 foot)

9

Idina can type 90 words in 2.5 minutes. How many words can she type in 12 minutes?

10

A salesman at a tea company makes a $15 commission on every $100 worth of products that he sells. If a jar of tea leaves is $20, how many jars would he have to sell to make $180 in commission?

11

A train covers 32 kilometers in 14.5 minutes. If it continues to travel at the same rate, which of the following is closest to the distance it will travel in 2 hours?

A) 54 kilometers
B) 265 kilometers
C) 364 kilometers
D) 928 kilometers

12

Peanut oil leaks out of an industrial container at the rate of 3 liters in 2 hours. If the peanut oil costs 8 dollars per liter, how many dollars' worth will be lost in 11 hours?

A) $60
B) $96
C) $118
D) $132

13

A recipe for soap calls for $1\frac{1}{2}$ cups of lye for every $\frac{2}{5}$ cup of castor oil. How many cups of lye are needed for a batch of soap that uses 3 cups of castor oil?

A) $1\frac{4}{5}$

B) 5

C) $9\frac{4}{5}$

D) $11\frac{1}{4}$

14

An 8 inch by 10 inch piece of cardboard costs $2.00. If the cost of a piece of cardboard is proportional to its area, what is the cost of a piece of cardboard that is 16 inches by 20 inches?

A) $4.00

B) $8.00

C) $12.00

D) $16.00

15

$$9 \text{ pikol} = 2 \text{ large bahar}$$
$$400 \text{ kulack} = 29 \text{ pikol}$$

The formulas above represent the relationships between some units of weight that were once used in Indonesia. A weight of 1,000 kulack is equivalent to how many large bahar? (Round your answer to the nearest whole number.)

16

Henry drives 150 miles at 30 miles per hour and then another 200 miles at 50 miles per hour. What was his average speed, in miles per hour, for the entire journey, to the nearest hundredth?

A) 38.89

B) 40.00

C) 42.33

D) 43.58

17

A "slow" clock falls behind at the same rate every hour. It is set to the correct time at 4:00 AM. When the clock shows 5:00 AM the same day, the correct time is 5:08 AM. When the clock shows 10:30 AM that day, what is the correct time?

A) 11:02 AM

B) 11:18 AM

C) 11:22 AM

D) 12:18 PM

18

Jared and Robert are accountants who are tasked with reviewing financial reports. It takes Jared 15 hours, working at a constant rate, to review a report containing 240 pages of financial statements. If Robert works at twice Jared's rate, how many minutes would it take Robert to review a report containing 120 pages of financial statements?

A) 225

B) 345

C) 450

D) 900

19

A flask contains an acidic solution with a concentration of 7.1×10^{15} hydrogen ions per milliliter. If 4.8×10^{23} hydrogen ions have a total mass of 0.8 grams, which of the following is closest to the concentration, in grams per liter, of the acidic solution?

A) 1.2×10^{-5}

B) 1.2×10^{-8}

C) 1.5×10^{-5}

D) 1.5×10^{-8}

20

Brett currently spends \$160 each month on gas. His current car is able to travel 30 miles per gallon of gas. He decides to switch his current car for a new car that is able to travel 40 miles per gallon of gas. Assuming the price of gas stays the same, how much will he spend on gas each month with the new car?

A) \$100

B) \$120

C) \$130

D) \$140

21

Margaret can buy 4 jars of honey for 9 dollars, and she can sell 3 jars of honey for 15 dollars. How many jars of honey would she have to buy and then sell to make a total profit of 132 dollars?

Chapter 4: Rates

CHAPTER EXERCISE:

1. $\boxed{14}$ For one week, Tim's diet plan would require a protein intake of $7 \times 60 = 420$ grams. Since each protein bar provides 30 grams of protein, he would need to buy $420 \div 30 = 14$ protein bars.

2. $\boxed{A}$ Over 6 years, the screen size increased by a total of $18.5 - 15.5 = 3$ inches. That's $3 \div 6 = 0.5$ inches each year.

3. $\boxed{B}$ The pressure increases by $70 - 50 = 20$ atm while the submarine descends $-900 - (-700) = -200$ meters. That's $20 \div 200 = 0.1$ atm per meter, or 1 atm per 10 meters.

4. $\boxed{3}$ The pool has a capacity of $5 \times 300 = 1,500$ gallons. At an increased rate of 500 gallons per hour, it would only take $1,500 \div 500 = 3$ hours to fill the pool.

5. $\boxed{B}$

$$20 \cancel{\text{apples}} \times \frac{d \text{ dollars}}{a \cancel{\text{apples}}} = \frac{20d}{a} \text{ dollars}$$

6. $\boxed{C}$ The racecar burned $22 - 18 = 4$ gallons of fuel in $7 - 4 = 3$ laps. To get to 6 gallons left, the racecar will have to consume $18 - 6 = 12$ more gallons. That's

$$12 \cancel{\text{gallons}} \times \frac{3 \text{ laps}}{4 \cancel{\text{gallons}}} = 9 \text{ more laps}$$

which is Lap $7 + 9 = 16$.

7. $\boxed{100}$ It took 2.5 hours for $65 - 40 = 25$ boxes to be unloaded. There are 3.5 hours from 3:30PM to 7:00PM. In 3.5 hours, $3.5 \cancel{\text{hours}} \times \frac{25 \text{ boxes}}{2.5 \cancel{\text{hours}}} = 35$ more boxes will be unloaded. That's a total of $65 + 35 = 100$ boxes.

8. $\boxed{120}$ Average speed is just total distance over total time. The total distance, in inches, was $2400 \times 12 = 28,800$. The total time, in seconds, was $4 \times 60 = 240$. $28,800 \div 240 = 120$ inches per second.

9. $\boxed{432}$

$$12 \cancel{\text{minutes}} \times \frac{90 \text{ words}}{2.5 \cancel{\text{minutes}}} = 432 \text{ words}$$

10. $\boxed{60}$

$$180 \cancel{\text{in commission}} \times \frac{100 \cancel{\text{in products}}}{15 \cancel{\text{in commission}}} \times \frac{1 \text{ jar}}{20 \cancel{\text{in products}}} = 60 \text{ jars}$$

11. $\boxed{B}$

$$2 \cancel{\text{hours}} \times \frac{60 \cancel{\text{minutes}}}{1 \cancel{\text{hour}}} \times \frac{32 \text{ kilometers}}{14.5 \cancel{\text{minutes}}} \approx 265 \text{ kilometers}$$

12. $\boxed{D}$ $11 \text{ hours} \times \frac{3 \text{ liters}}{2 \text{ hours}} \times \frac{8 \text{ dollars}}{1 \text{ liter}} = 132 \text{ dollars}$

13. [D] $3 \text{ cups of castor oil} \times \dfrac{\frac{3}{2} \text{ cups of lye}}{\frac{2}{5} \text{ cup of castor oil}} = 11.25 \text{ cups of lye}$

14. [B] An 8 inch by 10 inch piece of cardboard has an area of $8 \times 10 = 80$ square inches. A 16 inch by 20 inch piece of cardboard has an area of $16 \times 20 = 320$ square inches.

$$320 \text{ square inches} \times \frac{2 \text{ dollars}}{80 \text{ square inches}} = 8 \text{ dollars}$$

15. [16] $1{,}000 \text{ kulack} \times \dfrac{29 \text{ pikol}}{400 \text{ kulack}} \times \dfrac{2 \text{ large bahar}}{9 \text{ pikol}} \approx 16 \text{ large bahar}$

16. [A] The first 150 miles took $150 \div 30 = 5$ hours. The next 200 miles took $200 \div 50 = 4$ hours. His average speed, total distance over total time, was $(150 + 200)/(5 + 4) \approx 38.89$ miles per hour.

17. [C] The clock falls behind by 8 minutes every hour. There are 6.5 hours between 4:00 AM and 10:30 AM, so the clock falls behind by $8 \times 6.5 = 52$ minutes. The correct time is then 52 minutes past 10:30 AM, which is 11:22 AM.

18. [A] Jared's rate is $\dfrac{240}{15} = 16$ pages per hour. Robert's rate must then be $16 \times 2 = 32$ pages per hour. It would take Robert $\dfrac{120}{32} = 3.75$ hours to review the 120-page report. That's $3.75 \times 60 = 225$ minutes.

19. [A] $\dfrac{7.1 \times 10^{15} \text{ hydrogen ions}}{1 \text{ mL}} \times \dfrac{0.8 \text{ grams}}{4.8 \times 10^{23} \text{ hydrogen ions}} \times \dfrac{1000 \text{ mL}}{1 \text{ L}} \approx 1.2 \times 10^{-5} \dfrac{\text{grams}}{\text{L}}$

20. [B] What makes this question a little tricky is that we don't know the distance Brett travels each month or the number of gallons he uses each month. But we need to start somewhere, so let's say he needs 2 gallons of gas each month (you can make up any number you want). That means he travels $30 \times 2 = 60$ miles each month and each gallon costs $160 \div 2 = 80$ dollars (ridiculous, I know). Now if he switches to the new car, he'll only need $60 \div 40 = 1.5$ gallons of gas each month (distance of 60 miles divided by the 40 miles per gallon). Because the price of gas stays the same, that amount of gas will cost him $1.5 \times 80 = 120$ dollars each month. The answer ends up being 120 no matter what number we make up for the number of gallons of gas Brett uses each month.

    Here's an alternative solution. If we let x be the price per gallon of gas, then Brett currently uses $\dfrac{160}{x}$ gallons of gas each month. Using that amount of gas means that he drives $\dfrac{160}{x}(30) = \dfrac{4{,}800}{x}$ miles each month. With the new car, he will need $\dfrac{4{,}800}{x} \div 40 = \dfrac{120}{x}$ gallons of gas each month to drive that distance. Since each gallon of gas costs x dollars, he will need to spend $\dfrac{120}{x} \times x = 120$ dollars on gas each month.

21. [48] Each jar of honey costs $9 \div 4 = 2.25$ dollars. She can sell each jar for $15 \div 3 = 5$ dollars. That's a profit of $5 - 2.25 = 2.75$ dollars per jar. To make a profit of 132 dollars, she would have to sell $132 \div 2.75 = 48$ jars.